



Multi-model verification and evaluation of precipitation forecasts and Southwest China Vortex Rainstorms in Eastern Tibetan Plateau

Zhipeng Wu^{a,b}, Yan Zhang^{a,*}, Chunmei Hu^a, Chengzhi Deng^a, Juan Luo^a, Shenggang Wu^a, Yingying Zhou^a, Hong Zhang^a, Zhiyi Liao^a, Huan Wang^a, Yue Pang^a

^a Chongqing Meteorological Observatory, Chongqing, China

^b Nanchuan District Meteorological Observatory, Chongqing, China

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ABSTRACT

This study systematically evaluates the performance of five operational numerical models (CMA-GFS, ECMWF, NCEP-GFS, CMA-MESO, and CMA-SH9) during the 2021 flood season in the eastern Tibetan Plateau. Combining traditional verification metrics (Threat Score score, false alarm ratio, missing ratio) with subjective synoptic evaluation by forecasters, we assess model performance across seasons, months, and weather types, with particular focus on 22 severe rainfall events. Key findings include: 1) ECMWF demonstrates superior performance for light-to-moderate precipitation, while CMA-MESO and CMA-SH9 show better skill in heavy rainfall prediction, with assimilation systems significantly improving short-term (≤ 12 h) forecasts; 2) compared to its global counterpart, CMA-MESO demonstrates notably reduced surface temperature errors through its upgraded land data assimilation system.; 3) CMA-SH9 and CMA-MESO outperform ECMWF in positioning spring heavy rainfall events and capturing convective-scale features, though CMA-MESO's higher false alarm rates reduce its objective scores; 4) for frontal systems and southwest vortex events with cold air involvement, CMA-MESO provides the most accurate convective initiation locations and echo patterns.

1. Introduction

Heavy rainfall is one of the most significant meteorological disasters in China. Despite substantial improvements in infrastructure over recent decades, frequent rainstorm-induced disasters continue to cause considerable casualties and economic losses (Zechun et al., 2019). The Sichuan-Chongqing Basin, located on the eastern edge of the Tibetan Plateau, is influenced by multi-scale weather systems under the Asian monsoon during the flood season. Its complex terrain and intricate river systems contribute to a complex mechanisms for heavy rainfall and severe convective weather (Li Guoping, 2002). Additionally, this region remains one of the most challenging areas for numerical weather prediction (Yueqing, 2000). Precipitation in the eastern part of the basin is affected by numerous factors, leading to diverse heavy rainfall mechanisms, including frontal rainstorms triggered by cold air intrusions, low-level jet-induced rainstorms dominated by warm-moist airflows, and vortex rainstorms associated with the development of the Southwest China Vortex (SWV; Yueqing et al., 2021 and Xin et al., 2023). Further complicated by topographic forcing, each case exhibits unique

characteristics (Qiang et al., 2020; Chengzhi et al., 2021). At the onset of the flood season, in addition to synoptic-scale systems, meso- and micro-scale convective systems can also produce extreme rainfall, often with sudden onset, rapid movement, intense weather phenomena, and greater destructive potential (Bo et al., 2017).

The diversity of operational forecasting models has been increasing, and forecast decisions now typically integrate synoptic-scale, mesoscale, and ensemble prediction models, each with distinct features that require systematic evaluation. Casati et al. (2008, 2022) systematically reviewed the application principles of verification and evaluation methods across diverse scenarios, highlighting that research is shifting toward more meaningful verification approaches to address specific user needs, including new strategies for forecast types and operational optimizations. Xiaofeng and Rongwei (2021) compared the performance of European Centre for Medium-Range Weather Forecasts - Integrated Forecasting System (ECMWF-IFS), National Centers for Environmental Prediction - Global Forecast System (NCEP-GFS), and China Meteorological Administration - Global Forecast System (CMA-GFS) in forecasting precipitation over the Yalong River Basin in 2018, noting that

* Corresponding author at: Chongqing Meteorological Observatory, Chongqing, China.

E-mail addresses: 361913145@qq.com, 751246145@qq.com (Y. Zhang).

synoptic-scale models exhibit limited skill in predicting moderate-to-heavy rainfall, with significant false alarms and misses. Tong et al. (2019) assessed the China Meteorological Administration - Shanghai 9 km Regional Model(CMA-SH9) and it's precipitation forecasts in southwestern China using standard verification metrics, the Extremal Dependency Index(EDI) method, and the Method for Object-based Diagnostic Evaluation(MODE) method. Their findings indicated that while the model could capture the spatial distribution of heavy rainfall induced by the Southwest Vortex, it exhibited intensity biases. Gofa et al. (2018) simulated an intense convective event in the Mediterranean region using the Consortium for Small-scale Modeling (COSMO) model at two different resolutions, demonstrating that various spatial verification methods and averaging techniques can each provide distinct insights into model performance. Jing et al. (2022) applied object-based and neighborhood verification methods to evaluate CMA-SH9, China Meteorological Administration - Mesoscale Numerical Prediction System (CMA-MESO), and China Meteorological Administration - Data Assimilation and Modeling Branch System (CMA-DB) models during the 2019 flood season in Liaoning. By comparing false alarm rates, hit rates, spatial patterns, centroid displacements, and overlap between forecasts and observations, they found that forecast-observation consistency varied significantly depending on weather systems, with cyclone-induced precipitation showing ~10 % higher overlap than general rainfall, while post-high-pressure rainfall had higher miss rates. Zhipeng et al. (2021) evaluated Southwest Vortex-related precipitation over a year and found that all boundary layer parameterization schemes in models overestimated boundary layer height and turbulent mixing intensity. By adjusting model parameters based on observations, they improved the simulation of thermal and moisture structures, enhancing heavy rainfall prediction accuracy while reducing false alarms. Qing et al. (2020) comprehensively assessed national gridded forecast products and provincial-level corrected forecasts for precipitation and temperature from 2016 to 2018, comparing them with CMA and ECMWF outputs. Their results demonstrated a progressive increase in forecast value from raw model outputs to national guidance and further to localized corrections, forming a virtuous cycle in operational forecasting.

These studies have deepened the understanding of model biases across regions and laid the foundation for bias correction from multiple perspectives. Clearly, systematic and region-specific model evaluation not only supports forecast decision-making and improves public weather services but also provides feedback for model refinement, driving further advancements.

According to precipitation statistics from 34 counties/districts in eastern Sichuan-Chongqing Basin compiled by the Chongqing Climate Center, the results show that 2021 ranked as the fourth wettest flood season in the region since 1961, with the highest frequency and accumulated rainfall in recent years. This study evaluates five operational models during the 2021 flood season, employing verification algorithms and synoptic assessment by forecasters. We systematically analyze their precipitation forecast performance across seasons, months, and weather types. Furthermore, we assess model characteristics in 22 high-impact weather events dominated by the Southwest Vortex, cold air, and shear lines, revealing systematic biases in operational forecasts for the eastern Sichuan-Chongqing Basin.

2. Data and methods

2.1. Data and case selection

To comprehensively evaluate forecasts of general precipitation, characteristics of severe convection, and heavy rainfall events(In the red box area of Fig. 1, if 20 % of the rainfall stations recorded 24-h precipitation ≥50 mm; otherwise, it is considered general precipitation.). This study examines the period from 30 March to 18 September 2021 (totaling 173 days), spanning from the first severe convective weather

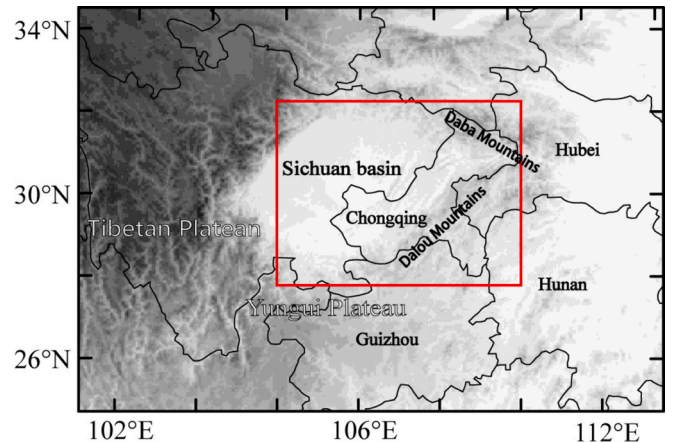


Fig. 1. Schematic diagram of mountain terrain and verification area (within the red box). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

event to the final regional rainstorm episode. A total of 22 high-impact weather cases were selected for analysis, with all timestamps reported in Beijing Time (UTC + 8).

Five operational NWP models were utilized, configuration information see Table 1. Three Global models: ECMWF-IFS, NCEP-GFS, CMA-GFS, and two regional models: CMA-SH9 (East China regional model, primarily used for high-resolution forecasting over the Yangtze River Delta region), CMA-MESO (focusing particularly on nowcasting of high-impact weather). Note that following October 2021, the original GRAPES (Global/Regional Assimilation and Prediction System) model was officially renamed as the CMA (China Meteorological Administration) Model. Its mesoscale version, CMA-MESO, expanded its coverage to 70–145°E and 10–60°N, encompassing nearly all of East Asia, and its forecast range was extended from 36 to 48 h. For consistency in this study, only the first 36-h forecasts from CMA-MESO were evaluated.

Observational Data includes:158 national meteorological surface stations (including basic, reference, standard and quality-controlled regional stations), contains standard components across the eastern basin. Gridded precipitation Data: The CMA Multi-Source (Ground-Satellite-Radar) Merged Precipitation Analysis Product (0.05° × 0.05° resolution over China), developed by the National Meteorological Information Center, was employed for precipitation analysis in the red box (Fig. 1). Radar data: Composite reflectivity mosaics were generated from 20 radars covering the eastern Sichuan-Chongqing Basin and adjacent areas.

2.2. Verification and evaluation methods

2.2.1. Objective scoring methods

The objective precipitation verification employs the Threat score (TS), False Alarm Ratio (FAR), and Missing Ratio (MR) as metrics,

Table 1
Basic configuration information of five models in this study.

Model	Prediction domain	Horizontal resolution	Vertical levels	Update cycle (per day)	Assimilation system
ECMWF	Global	12.5 km	137	2	4D-Var
NCEP-GFS	Globa	25 km	64	4	Hybrid 4D-EnVar
CMA-GFS	Global	25 km	78	2	CMA-4DVar
CMA-MESO	70-145E/ 10-60 N	3 km	50	8	3DVar/ 4DVar
CMA-SH9	90-150E/ 10-60 N	9 km	50	2	3DVar

calculated variables are listed in Table 2. Precipitation intensity is classified according to the China Meteorological Administration's 24-h precipitation grading standards: light rain (≥ 0.1 mm), moderate rain (≥ 10 mm), heavy rain (≥ 25 mm), and torrential rain (≥ 50 mm)

a. represents hits (correct forecasts), b. indicates false alarms, c. denotes misses, and d corresponds to cases with neither precipitation observed nor forecasted.

TS (Threat Score) = $a/(a + b + c)$, score ranges from 0 to 1, with higher values indicating better forecast performance. FAR (False Alarm Ratio) = $b/(a + b)$, ranges from 0 to 1, with a score of 0 representing optimal forecast performance. It measures the proportion of forecasted events that did not actually occur. MR (Missing Ratio) = $c/(a + c)$, ranges from 0 to 1, with a score of 0 representing optimal forecast performance. It measures the proportion of observed events that were not forecasted. For long-term verification results, the counts of events a, b, c, and d are summed over the specified time period before calculating these metrics.

Root Mean Square Error (RMSE) is calculated as follows:

$$RMSE = \frac{1}{N} \sum_{i=1}^N (F_i - O_i)^2 \quad (1)$$

where i represents the station index for verification, N is the total number of stations. The Root Mean Square Error (RMSE) is used to quantify the deviation between observed values and true values. The RMSE over a specified time period represents the average of RMSE values within that period.

For the Distribution of Amount with hourly Intensity (DAI), which examines the model's capability to forecast precipitation amounts across different hourly intensity categories, the following two-parameter exponential equation is employed to fit the precipitation intensity distribution:

$$A(I) + 1 = e^{\alpha - \frac{1}{\beta} I} \quad (2)$$

where I represents hourly precipitation intensity, A(I) denotes the total precipitation amount corresponding to intensity I, and α and β are two fitting parameters that characterize the quantitative distribution of precipitation with intensity. The closer the modeled curve is to the observed curve, the better the model's precipitation-amount-intensity structure matches reality.

2.2.2. Synoptic evaluation

As precipitation forecasts become more refined and higher in magnitude, achieving high objective verification scores becomes increasingly challenging. If model-predicted precipitation areas are adjacent to but slightly displaced from observed areas, the forecast may still provide valuable decision-making information to human forecasters. However, the verification algorithm will impose a "double penalty" (simultaneous false alarms and misses) in such regions, resulting in artificially low objective scores. Incorporating subjective evaluation by meteorologists can effectively extract the value of high-resolution numerical forecasts in these cases.

For the synoptic evaluation, this study engaged 12 provincial forecasters (including 4 chief forecasters) to conduct a subjective assessment of model performance. They analyzed 22 heavy rainfall cases listed in Table 3, evaluating forecasts from 5 models based on five key aspects: moisture transport, wind fields, thermal conditions, convective characteristics, and precipitation patterns. Since most evaluators had

Table 3

High-impact precipitation cases and their major affected weather systems in Eastern part of the Sichuan-Chongqing Basin in 2021.

Case	Primary impact period	Dominant synoptic system	Maximum accumulated & Peak hourly rainfall (mm)
Mar 30	2000 Mar 29–2000 Mar 30	SWV	139.4 /53.8
Apr 24	2000 Apr 23–0800 Apr 24	Shear	161.9 /29.7
May 02	2000 May 02–0800 May 03	Front	249.1 /138.5
May 14	2000 May 13–2000 May 14	Shear	124.7 /68.4
Jun 17	2000 Jun 17–0800 Jun 18	SWV	323.7 /78.9
Jun 26	2000 Jun 26–0800 Jun 27	Shear	319.4 /121.3
Jun 27	2000 Jun 27–2000 Jun 28	SWV	319.4 /121.3
Jul 06	2000 Jul 06–1400 Jul 07	SWV	288.6 /92.8
Jul 07	2000 Jul 07–0800 Jul 08	Shear	288.6 /92.8
Aug 07 Episode I	0800 Aug 07–0800 Aug 08	SWV	261.2 /84.7
Aug 07 Episode II	0800 Aug 08–0800 Aug 09	SWV	257.4 /85.9
Aug 12	2000 Aug 12–2000 Aug 13	SWV	283.3 /81.2
Aug 22	2000 Aug 22–2000 Aug 23	SWV	225.8/86.4
Aug 25 Episode I	2000 Aug 25–2000 Aug 26	Front	167.7/34.7
Aug 25 Episode II	0800 Aug 28–2000 Aug 28	Shear	112.4 /19.3
Aug 25 Episode III	2000 Aug 28–0800 Aug 29	Shear	166.8.1 /40.4
Sep 04	2000 Sep 04–0800 Sep 05	Front/SWV	300 /84.0
Sep 06	2000 Sep 05–2000 Sep 06	Shear	212.1 /83.9
Sep 11	2000 Sep 11–0800 Sep 12	SWV	157.5 /108.4
Sep 15	2000 Sep 15–2000 Sep 16	Front /Shear	290 /78.1
Sep 18	2000 Sep 18–2000 Sep 19	Front	127.8 /48

personally forecasted these events, they were given full discretion in their assessment. While maintaining synoptic analysis rigor, each forecaster ultimately scored each event on a 0–10 scale: 0 indicates no reference value or even misleading for decision-making; a score of 6 show forecasts with passable reference value for precipitation location, morphology, and intensity; 10 represents model result matches operational forecasting conclusions.

The final evaluation score for each model was obtained by:1. Averaging scores across all 22 cases to eliminate individual biases, 2. Integrating analysis conclusions from heavy rainfall cases. This process yielded comprehensive assessment scores for all 5 models.

Table 2

standard 2 × 2 bicategorical event contingency table.

Observation/Forecast	Forecast Yes	Forecast No
Observed Yes	a	c
Observed No	b	d

3. Verification and evaluation results

3.1. Overall verification during flood season

Figures 2 and 3 present the TS scores of long-term verification for five models from 30 March to 18 September (totaling 173 days) during the flood season, initialized at 08:00 and 20:00. The ECMWF model generally exhibits better and more stable performance for precipitation forecasts below moderate rain, followed by the CMA-MESO and CMA-SH9 model. In the first 6 h after initialization, the CMA-MESO and CMA-SH9 models outperform other models in TS scores for most precipitation thresholds, which is likely attributed to radar data assimilation. However, this assimilation-enhanced advantage diminishes within approximately 12 h, consistent with the findings of Liping et al. (2022), refer to Fig. 4. Although the 24-h cumulative precipitation TS scores of CMA-MESO are slightly lower overall than those of the ECMWF model, the 3-hourly scores—particularly for higher precipitation thresholds—are significantly better than ECMWF (Fig. 3d–e).

For heavy rain or above, mesoscale models demonstrate a clear advantage over global models, which is associated with their superior capability in forecasting high-intensity precipitation. The CMA-MESO model achieves TS scores exceeding 0.1 for heavy rain in the first 6 h (Fig. 2e, 3e), outperforming CMA-SH9.

In 2021, the CMA-MESO model underwent an upgrade, significantly improving its precipitation forecasting capability in the short-term (Fig. 4). Compared to the previous version, the main enhancements included improvements to the assimilation system, such as variational assimilation, land surface assimilation, and cloud analysis, along with the implementation of a series of radar quality control schemes. These upgrades notably enhanced the model's precipitation forecasting skill within the first 12 h, particularly in the short-term range (Liping et al., 2022).

For moderate precipitation and above, CMA-MESO and CMA-SH9 exhibit a clear advantage over ECMWF at around 12 h (Fig. 2–3, d–f).

However, this advantage diminishes with increasing forecast lead time. Taking CMA-MESO as an example, the benefits of incorporating radar quality control and cloud analysis into the assimilation system completely disappear by approximately 30 h. This suggests that in heavy precipitation forecasting services, mesoscale models should be primarily referenced for high-intensity precipitation and convective characteristics within the first half-day after initialization, while longer lead times provide much less value. Furthermore, initiating a rapid-cycling assimilation system at approximately 6-h intervals can help mitigate the decay of assimilation gains, ensuring continuous provision of convective cloud analysis information.

Based on the root-mean-square error (RMSE) and bias verification of 2 m temperature at 3-hourly intervals for each model (Fig. 5), the results from 08:00 and 20:00 initializations are generally consistent, with no apparent differences in bias among the models. The ECMWF model exhibits the smallest RMSE, CMA-MESO shows notable improvement after the upgrade of its land surface assimilation system, compared to its global vision, ranking second. CMA-SH9 and CMA-GFS perform similarly, whereas NCEP-GFS demonstrates relatively poorer performance. ECMWF maintains a clear advantage in 2 m temperature field forecasting.

The accuracy of a model's underlying surface significantly impacts its overall performance. The temperature, moisture, and latent heat flux of the soil directly or indirectly influence the thermal and moisture stratification in the upper levels through the model's boundary layer parameterization scheme. Since the atmospheric stratification state directly determines convective stability, the precision of the initial underlying surface conditions indirectly governs the accuracy of the atmospheric stratification. The land surface assimilation system provides the model with these initial surface conditions. Lili et al. (2018), referencing the ECMWF's land surface assimilation scheme, established probabilistic statistical relationships among soil moisture, 2 m temperature, and relative humidity. They implemented an optimal interpolation-based land surface assimilation scheme in CMA-MESO,

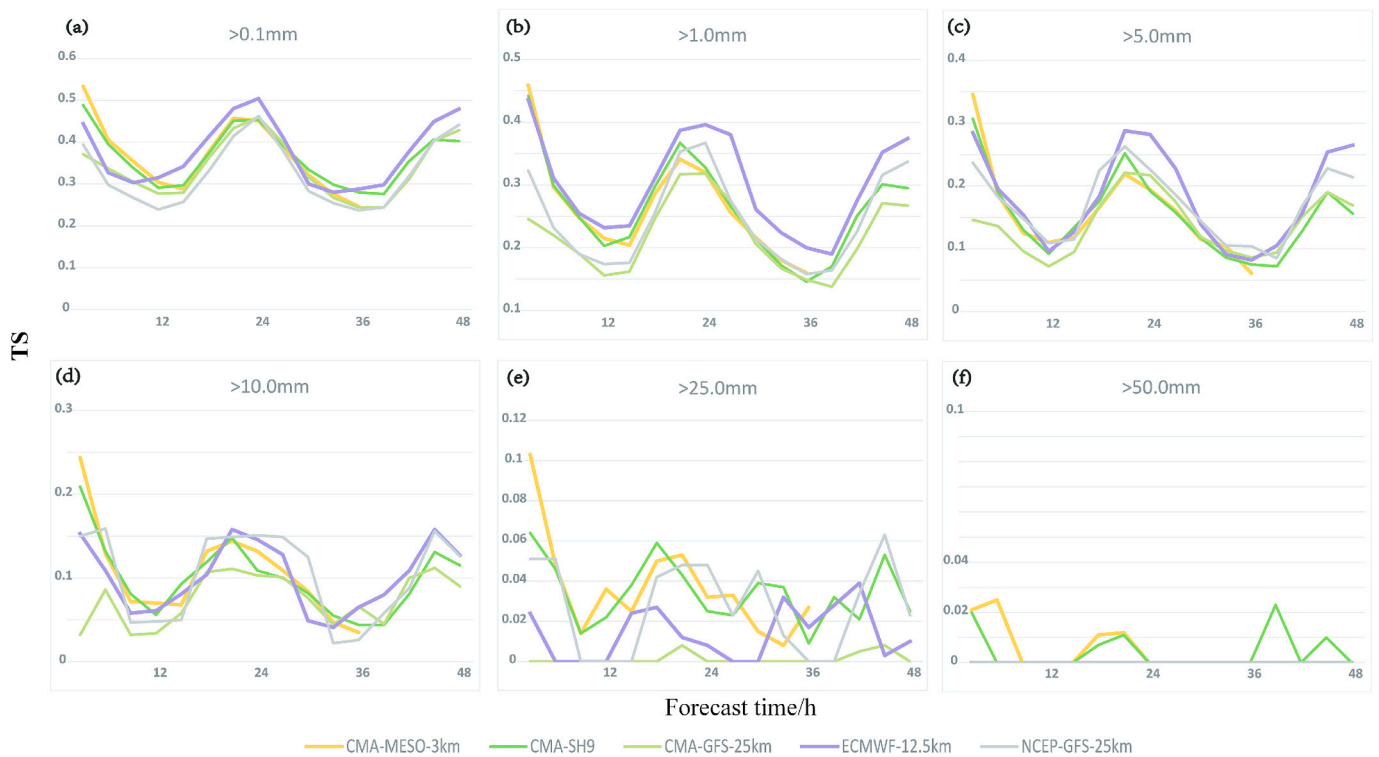


Fig. 2. TS scores of 3-h accumulated precipitation at varying thresholds over the eastern Tibetan Plateau from multiple models (initialization: 08:00 UTC, verification period: 173 days, forecast lead time: 48 h, March 30 to September 18, 2021 (a) Threshold > 0.1 mm, (b) Threshold > 1 mm, (c) Threshold > 5 mm, (d) Threshold > 10 mm, (e) Threshold > 25 mm, (f) Threshold > 50 mm.

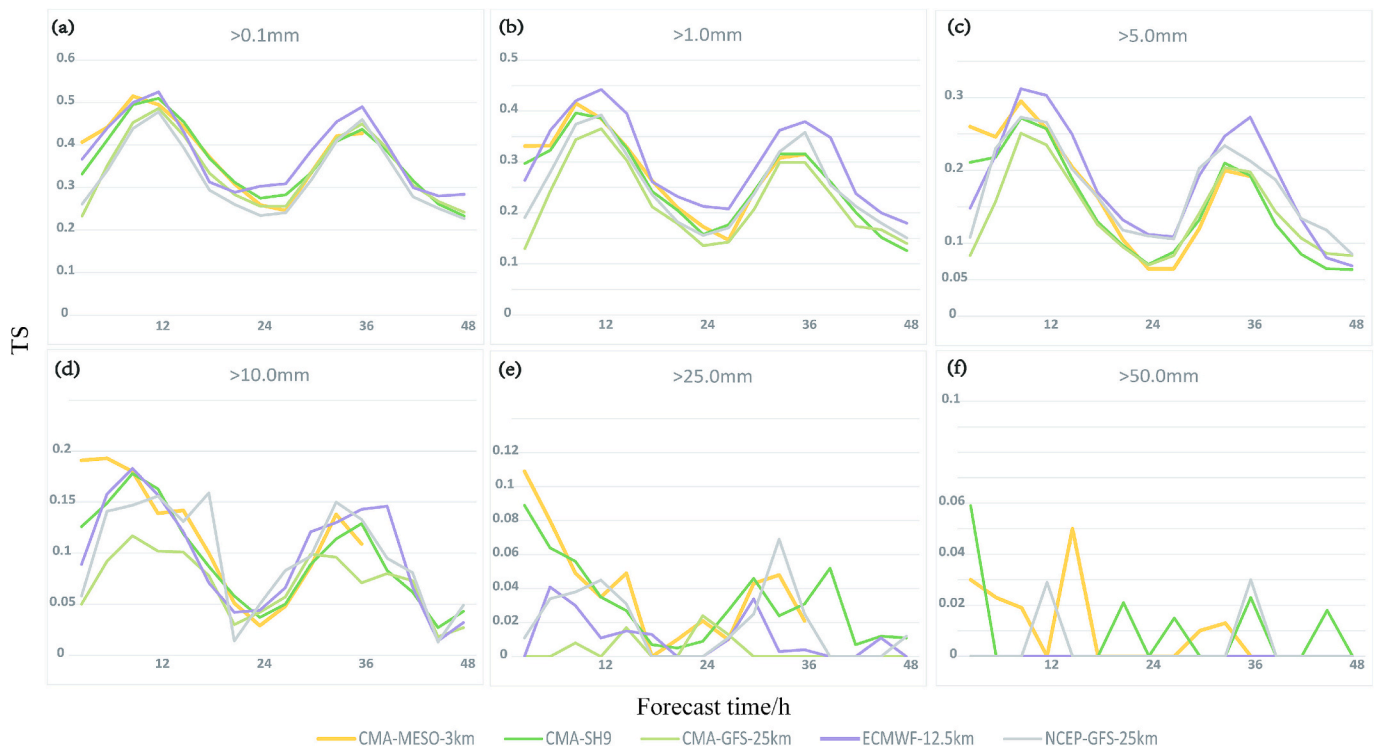


Fig. 3. TS scores of 3-h accumulated precipitation at varying thresholds over the eastern Tibetan Plateau from multiple models (initialization: 20:00 UTC, verification period: 173 days, forecast lead time: 48 h), March 30 to September 18, 2021 (a) Threshold > 0.1 mm, (b) Threshold > 1 mm, (c) Threshold > 5 mm, (d) Threshold > 10 mm, (e) Threshold > 25 mm, (f) Threshold > 50 mm.

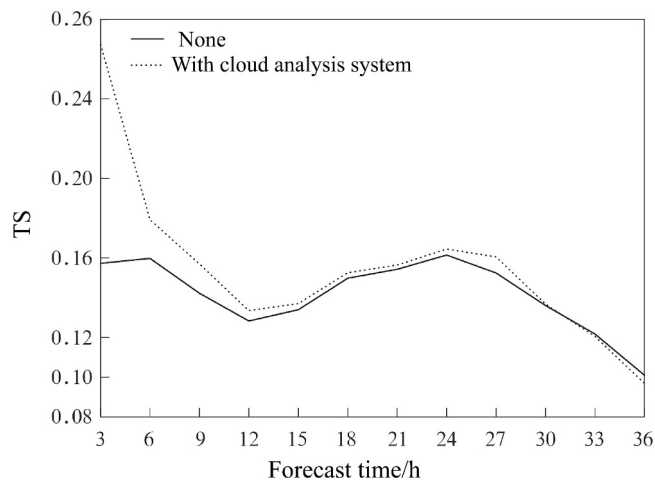


Fig. 4. TS Score for 3 h accumulated precipitation above 5 mm threshold from 1 Jun to 31 Aug in 2019 (As cited in Liping et al., 2022).

which notably reduced 2 m temperature forecast bias (Fig. 5c-d). Their study also demonstrated that this improvement enhanced 12-h now-casting of precipitation, a finding consistent with the results of this research.

3.2. Verification for spring and summer

Chongqing and the northeastern Sichuan cities of Guangan and Dazhou are located in the hilly terrain east of the Tibetan Plateau, characterized by complex topography and river systems. This makes it particularly challenging for models to accurately represent the underlying surface in this region. During spring, the alternation between dry cold air from the north and warm moist air from the south frequently

triggers meso- and micro-scale convection. Unlike the widespread heavy rainfall typical of summer, spring often brings scattered but intense convective weather. These events, though covering small areas (often just a few kilometers), can produce extremely heavy hourly rainfall (up to tens or even hundreds of millimeters, see Table 3) and are usually accompanied by thunderstorms, gusts, and hail. Models generally perform poorly in forecasting such meso- and micro-scale severe weather, with significant false alarms and misses, necessitating separate verification for spring and summer.

Figures 6 and 7 show the 24-h conventional verification results for the five models in spring and summer, initialized at 08:00 and 20:00, respectively. In terms of TS scores, ECMWF performs best overall, with relatively lower false alarm and miss rates compared to other models, and demonstrates clear advantages for precipitation above 5 mm. During spring, CMA-SH9 outperforms CMA-MESO, while the opposite is true in summer. Notably, CMA-MESO exhibits significant false alarm characteristics in spring. Although the false alarm rate decreases in summer, it remains relatively high. CMA-MESO tends to predict large, concentrated areas of heavy precipitation for spring convection, whereas the observation is more scattered, consistent with the false alarm results shown in Fig. 6b-e. Additionally, CMA model forecasts initialized at 08:00 show greater bias compared to those initialized at 20:00, with CMA-MESO's false alarm characteristics being more pronounced in the 08:00 runs. This inter-cycle discrepancy may primarily stem from: diurnal cycle-assimilation timing mismatches, model background fields quality or spin-up duration variations.

3.3. Verification and assessment of high-impact precipitation events

Climatic records from the Chongqing Climate Center indicate that 2021 registered the fourth-highest regional precipitation total since 1961 across 34 counties/districts in the eastern Sichuan-Chongqing Basin, with both frequency and accumulation extremes surpassing recent decadal averages. During these high-impact precipitation events,

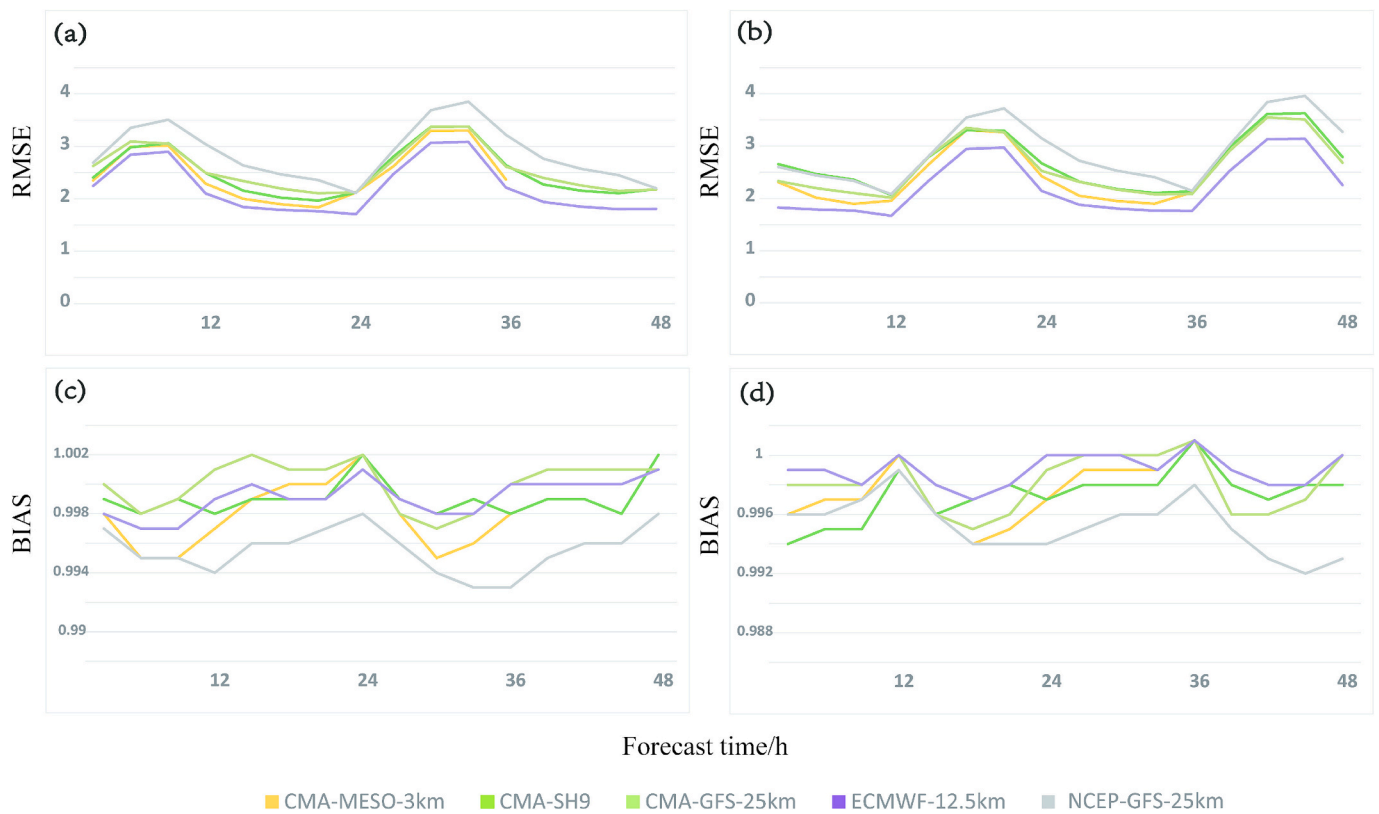


Fig. 5. Eastern part of Tibetan Plateau 2 m-temperature scores of verification forecast time at 08:00 and 20:00 from Mar 30 to Sep 18, 2021 for multi-mode (Test days: 173) (a) 2 m temperature Root-Mean-Square-Error at 08:00, (b) 2 m temperature Root-Mean-Square-Error at 20:00, (c) 2 m temperature bias at 08:00, and (d) 2 m temperature bias at 20:00.

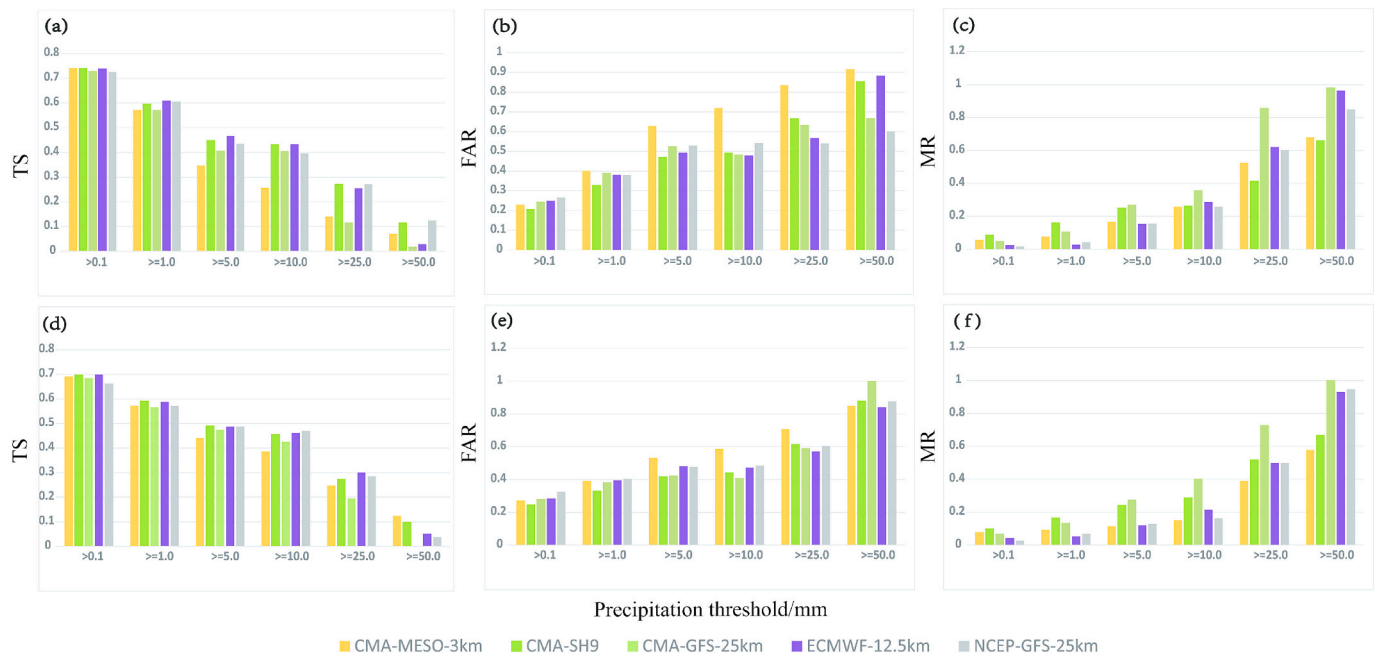


Fig. 6. 24-h precipitation verification scores for Multi-Model in eastern part of the Tibetan Plateau at different threshold from Mar 30 to May 14, Model initial time at 08:00 and 20:00 (Spring Inspection Days: 46) (a) TS Score at 08:00, (b) False-Alarm Rate at 08:00, (c) Missing Rate at 08:00, (d) TS Score at 20:00, (e) False-Alarm Rate at 20:00, (f) TS Score at 20:00.

the maximum accumulated rainfall reached 323.7 mm, with a peak hourly intensity of 138 mm. Some heavy rainfall events persisted for extended periods, accompanied by significant evolutions in the

influencing weather systems. Therefore, the events were analyzed in segments, totaling 22 cases.

The results demonstrate that precipitation events associated with the

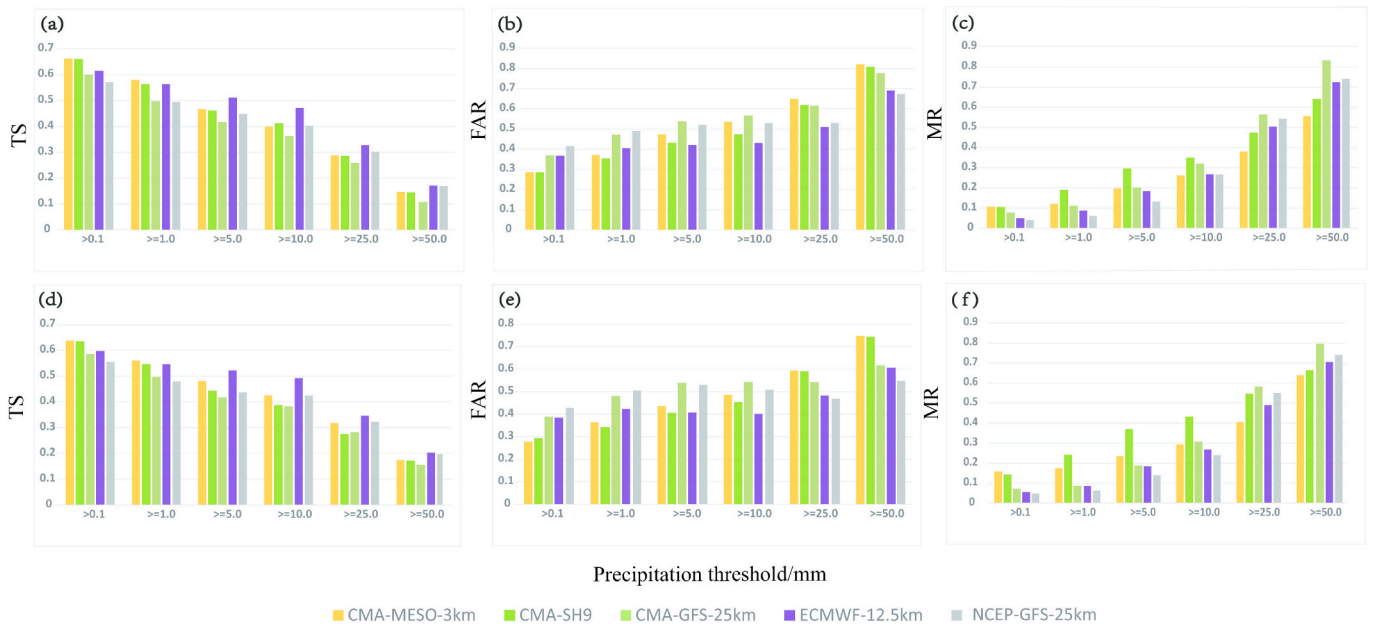


Fig. 7. 24-h precipitation verification scores for Multi-mode in Eastern part of the Tibetan Plateau at different threshold from May 15 to Sep 18, 2021, Model initial time at 08:00 and 20:00 (Spring Inspection Days: 46) (a) TS Score at 08:00, (b) False-Alarm Rate at 08:00, (c) Missing Rate at 08:00, (d) TS Score at 20:00, (e) False-Alarm Rate at 20:00, (f) TS Score at 20:00.

Southwest Vortex (SWV) exhibited particularly high disaster potential, characterized by both substantial accumulated rainfall and intense hourly precipitation rates. Although the primary influencing systems are highlighted in the table, subtle structural variations were observed, for instance, the “May 02” and “May 14” events still featured low-level vorticity convergence, though less dominant compared to the shear lines forced by cold air. The “Aug 25” event displayed a dual-vortex shear pattern steered by cold air, thus not classified as a typical SWV event.

These findings underscore the crucial role of the Southwest Vortex’s genesis, development, and eastward propagation in driving high-impact precipitation across the eastern Sichuan-Chongqing Basin. In particular, when strong cold air moves southward, causing a daily average temperature drop of 8 °C or more, the combined effects of the frontal system and the SWV may result in more severe hazardous weather impacts.

3.3.1. Objective scoring of cases

Based on the key cases listed in Table 3, the average TS scores for spring and summer are calculated (Fig. 8). In spring, the ECMWF model shows better performance for light to moderate rainfall (moderate rain and below), while CMA-SH9 outperforms other models for heavy rain and above. In summer, the ECMWF model achieves the highest TS scores for all precipitation levels below torrential rain, though it slightly underperforms mesoscale models at the torrential rain level. CMA-MESO exhibits significant improvement in summer compared to spring and generally performs better than CMA-SH9.

Figure 9 shows the distribution of hourly precipitation intensity forecasts from March to September, with the black curve representing observations. From March to Apr, the mesoscale models exhibit significantly overestimated hourly rainfall intensity, particularly in March and April. CMA-MESO’s 24-h precipitation intensity forecast is notably excessive, combined with its high false alarm ratio (Fig. 6b), which likely explains its poorest TS score for spring precipitation. In contrast,

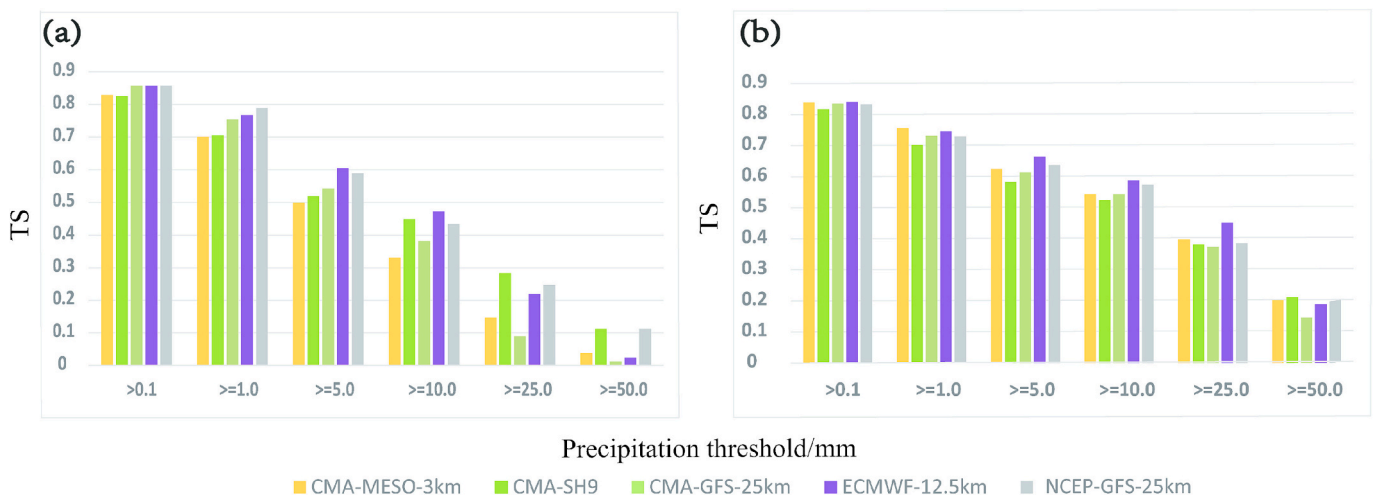


Fig. 8. TS scores for Multi-model of typical severe convection and heavy rain fall cases in spring and summer in Eastern part of the Tibetan Plateau, including 4 cases in spring and 18 cases in summer (a) TS score of spring cases, (b) TS score of summer cases.

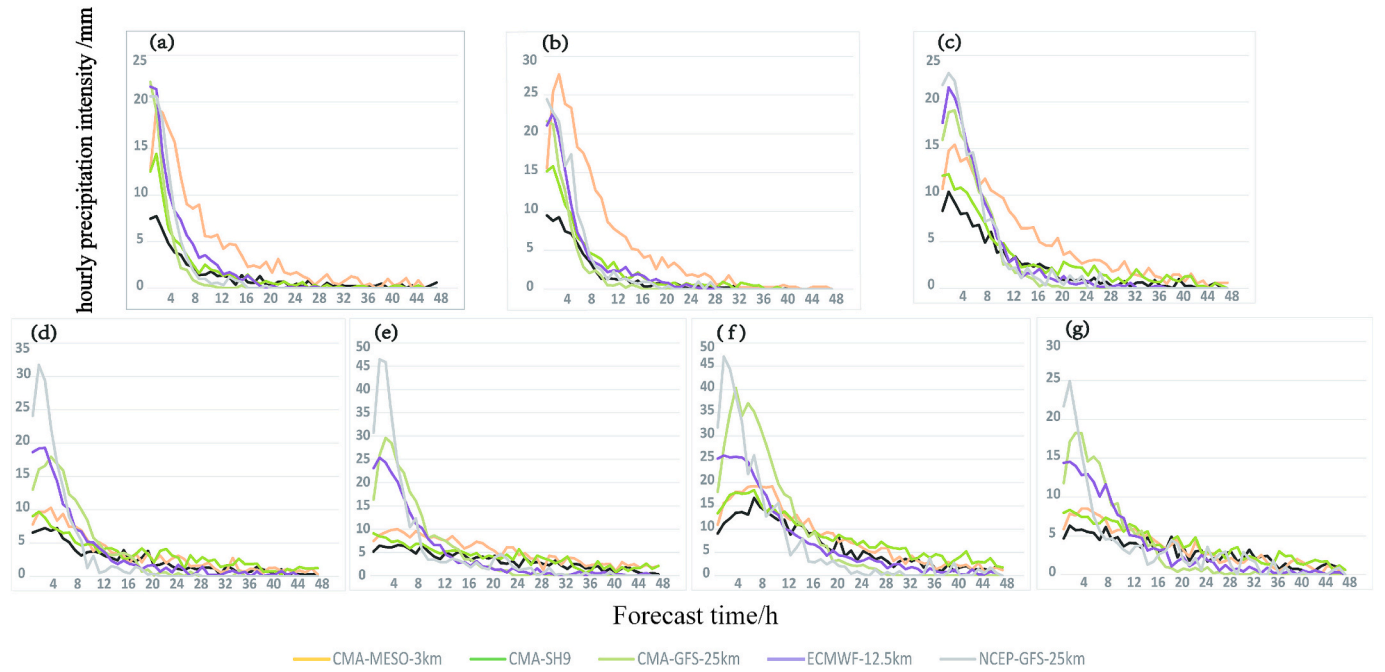


Fig. 9. Distribution characteristics of monthly precipitation with hourly intensity from March to September in eastern part of the Sichuan Basin Region (a) March, (b) April, (c) May, (d) June, (e) July, (f) August, (g) September.

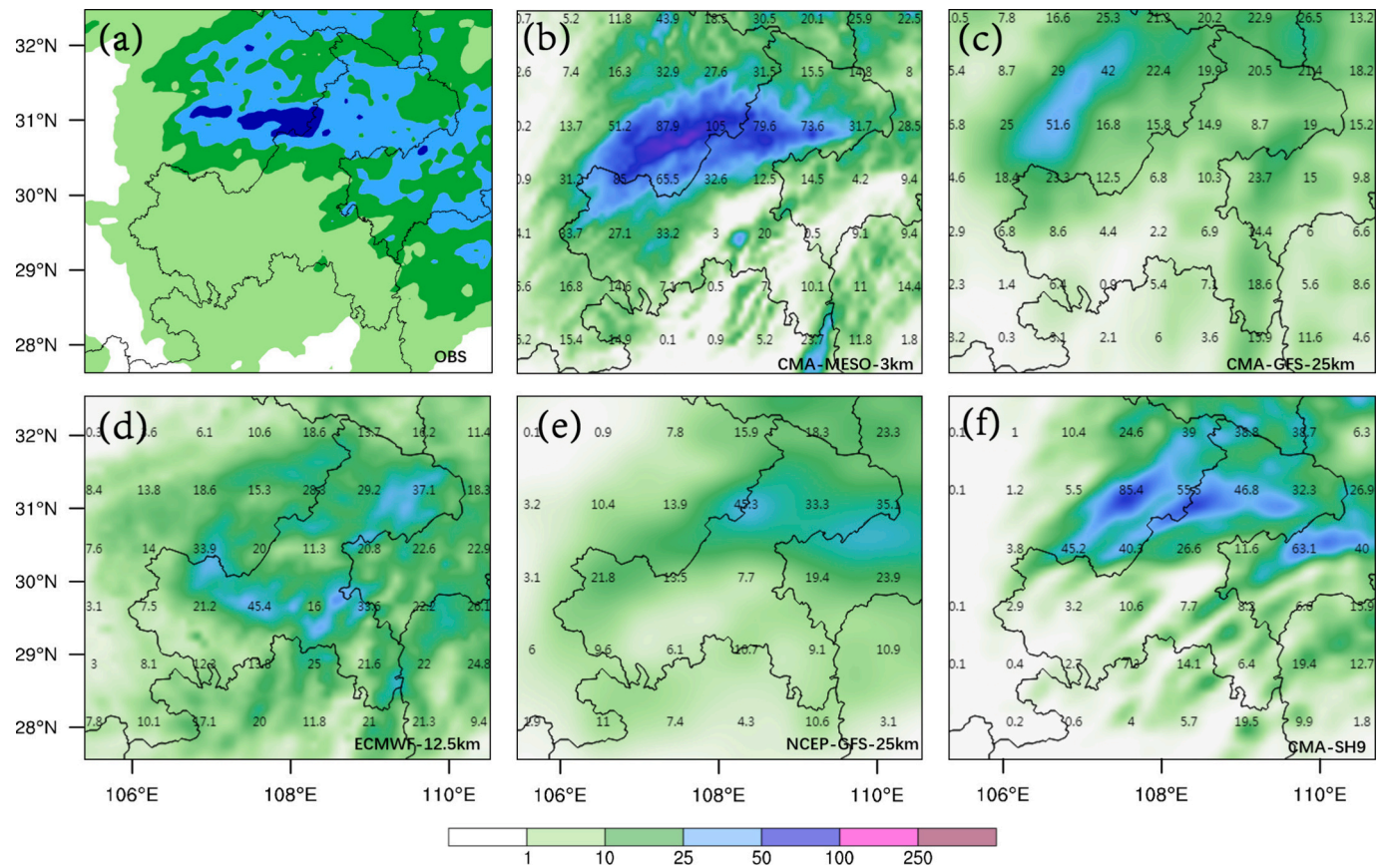


Fig. 10. Comparison of the observation precipitation and Multi-model forecast of the "Apr 24" Severe Convection case from 20:00 on April 23 to 08:00 on April 24, 2021 (a) Observation, (b) CMA-MESO, (c) CMA-GFS (d) ECMWF, (e) NCEP-GFS, (f) CMA-SH9.

(Note: The model initial time is all at 08:00 on April 23, 2021)

CMA-SH9's hourly rainfall intensity forecasts align most closely with observations. After June, as precipitation systems become more organized and coverage expands, the discrepancy between mesoscale model forecasts and observed hourly rainfall intensity diminishes, while the NCEP model demonstrates a pronounced overprediction of hourly precipitation intensity when the main flood season begins.

3.3.2. Synoptic evaluation by forecasters

From a purely precipitation perspective, the TS score integrates model performance characteristics including hit, false alarm and missed event rate, yet may obscure valuable forecast information. For instance, if mesoscale heavy rainfall forecasts are adjacent to but do not overlap with observed areas, the forecast would be penalized by both false alarms and missed events—even when the model provides valuable reference for forecast decision-making, the objective verification score would remain low.

Figure 10 compares observed precipitation with multi-model forecasts for the “Apr 24” case. Synoptic-scale models show significant divergence in forecasting this case: CMA-GFS predicted rainfall too far west, ECMWF underestimated intensity with excessive false alarms in southern regions, while NCEP-GFS performed best overall in capturing the heavy precipitation center and location, albeit with an under-predicted coverage. Both mesoscale models (CMA-MESO and CMA-SH9) successfully predicted the heavy rainfall areas in Chongqing and northeastern Sichuan, compensating for shortcomings in synoptic-scale models. However, CMA-MESO overestimated the spatial extent of heavy rainfall, producing notable false alarm regions, and exhibited large missed-event areas for heavy rain in the Daba Mountains and western Hubei. The increased false alarm and missed-event rates significantly reduced its TS score. Although both mesoscale models provided valuable forecasts—in predicting jet stream exit positions right at heavy

precipitation cores, and rainband morphology more consistent with observations—CMA-MESO's TS score was substantially lower than CMA-SH9's.

A significant low-level jet (LLJ) was observed at 700 hPa in the early morning of Apr 24. The location of the LLJ exit region in various models showed the most pronounced impact on heavy precipitation positioning. Notably, CMA-MESO simulated maximum jet speeds of 20–24 m/s, while other models generally produced weaker wind speeds of 16–20 m/s. CMA-MESO demonstrated valuable forecasting performance for the evolutionary characteristics of convective cloud clusters during this event. Its 6-h forecast maintained relatively accurate regarding the spatial distribution, morphology and the structure of convective clusters. However, the model exhibited progressively stronger intensity biases as forecast lead time increased (Fig. 11b), ultimately resulting in an elevated false alarm rate for precipitation predictions.

The current mesoscale modeling system has incorporated data assimilation from multiple radar networks. Taking CMA-MESO as an example, the system integrates quality-controlled data from over 200 radars (including SA/SB/CB/SC/CD/CC types), combined with Fengyun geostationary satellite products - Cloud Total Amount (CTA) and Black Body Brightness Temperature (TBB). Through the initial cloud analysis system, the model is initialized with comprehensive hydrometeor profiles including cloud water, rainwater, ice crystals, and snow particles, along with corresponding water vapor and potential temperature fields. These cloud initialization data are progressively introduced at the model integration onset, which has significantly enhanced nowcasting capability (at the first 6 h). The refined initialization enriches meso- γ -scale information in the initial fields with improved accuracy. Furthermore, the rapid-update-cycle assimilation system effectively preserves these convective-scale features throughout the cycling process (Bo et al., 2017).

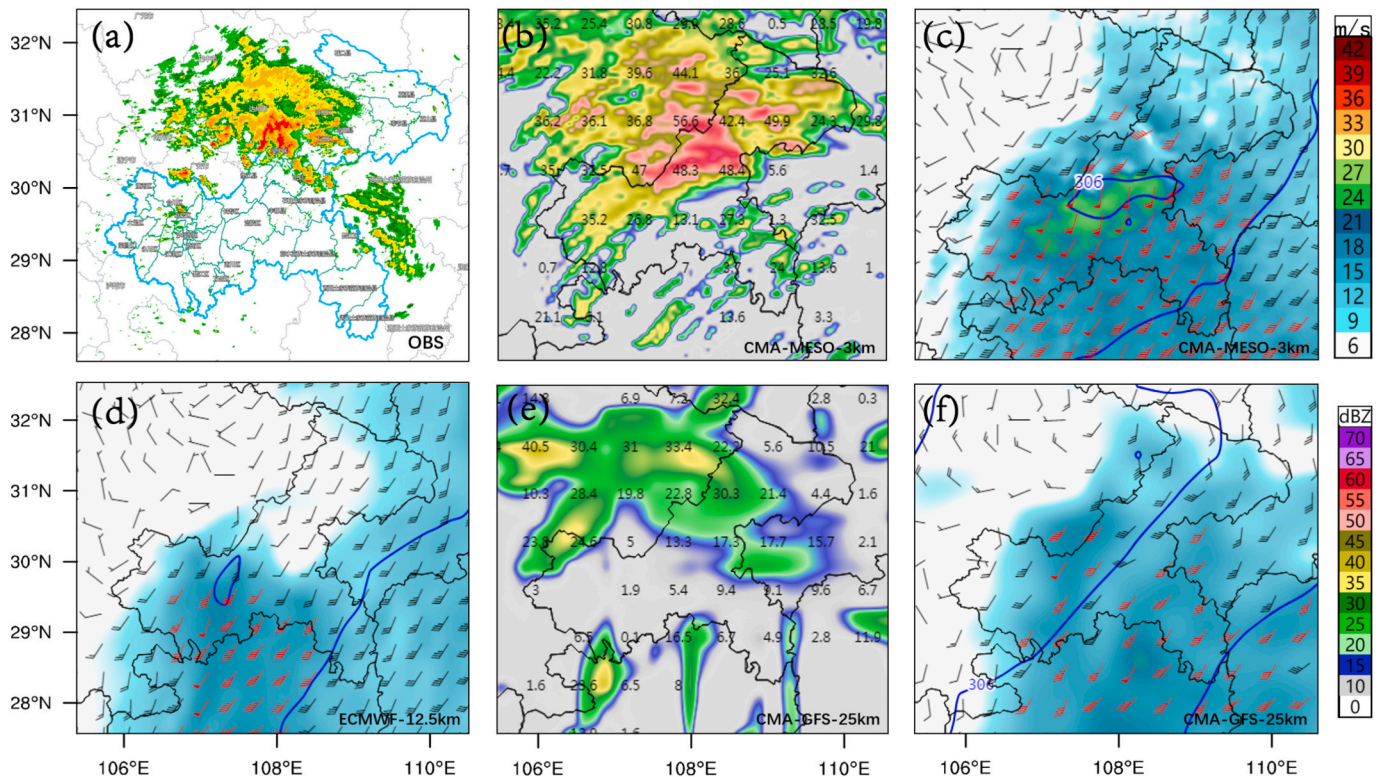


Fig. 11. On Apr 24, 2021 at 02:00, the radar echo (color shadow, unit: dbZ), wind field (wind scale and equal wind speed shadow, unit: m/s), and height field (contour line, unit: gpm) of the “Apr 24” severe convective weather cases were compared with the predictions: (a) observation of combined reflectivity, (b) CMA-MESO combined reflectivity prediction, (c) CMA-MESO 700 hPa wind field and height field prediction, (d) ECMWF 700 hPa wind and altitude field prediction, (e) CMA-GFS combined reflectance prediction, (f) CMA-GFS 700 hPa wind and altitude field prediction.

(Note: The model initial time is all at 20:00 on Apr 23, 2021)

The “Apr 24” case highlighted both the inherent limitations of global-scale models in simulating convective processes and the performance variations among mesoscale models. As shown in Fig. 8a, CMA-SH9 demonstrated superior performance for spring rainfall events when the precipitation threshold exceeds 25 mm, exhibiting notable skill in localization, morphology and intensity.

Although CMA-MESO also shows valuable information for heavy rainfall positioning and convective echo characteristics, its systematic biases—including stronger wind fields, amplified convective intensity, and excessive precipitation coverage among models—collectively degrade the TS core. The performance disparity between CMA-MESO and CMA-SH9 likely stems from fundamental differences in model physical parameterization or dynamic core formulation, however, detailed mechanistic analysis requires further investigation and falls beyond the scope of this discussion.

The “Sep 04” heavy rainfall case was a cold-air-influenced torrential rain process (Fig. 12). During the main flood season, the false alarm rate of CMA-MESO significantly decreased, demonstrating improved capability in predicting the rainband positions of heavy rainfall. Synoptic evaluations of multiple cases revealed that CMA-MESO exhibits better forecasting performance for cold-air-induced heavy rainfall processes, such as torrential rain cases on “Aug 25” and “Sep 15”, where it demonstrated relatively more accurate predictions of the southwest vortex and shear line positions compared to the evolution in the re-analysis data (Figure omitted for brevity). From the perspective of pressure fields, all models demonstrated relatively accurate forecasts. However, the mesoscale models exhibited superior capability in resolving stronger wind fields compared to synoptic-scale models. As shown in Table 3, during the cold-air-affected intense precipitation phase, The CMA-MESO showing cold air intrusion from the north with a more pronounced northerly jet stream compared to southerly winds. In

contrast, CMA-SH9 forecasts indicate a dominant southerly jet stream, exhibiting clear warm-sector precipitation characteristics with weaker northerly flows. Additionally, the CMA-MESO demonstrates relatively superior performance in forecasting the movement, structure, and position of the southwest vortex during the nocturnal process of the “Sep 04” case (Figure omitted for brevity).

To leverage forecasters’ subjective assessment capabilities from a synoptic analysis perspective, this study systematically collected and analyzed forecast-observation paired data for 22 cases listed in Table 3. The evaluation framework incorporated precipitation patterns, temperature fields, geopotential height fields, moisture fields, upper/lower-level wind fields, and composite radar reflectivity evolution maps. Twelve provincial forecasters with extensive operational warning experience conducted a comprehensive subjective synoptic evaluation. Through in-depth discussions, particular emphasis was placed on assessing forecast-observation discrepancies concerning southwest vortices, shear lines, and cold air intrusions.

Subjective Evaluation Scale 0–10 points. 0 (No operational value, even could mislead forecasting decisions); 6 (Moderate reference value, acceptable for precipitation location, pattern and intensity); 10 (Excellent consistency with forecasters’ operational decision-making). Note that, while inter-evaluator variability exists due to subjective criteria,

Table 4

Synoptic evaluation results of multi-model on 22 typical rainfall cases at the perspective from forecasters.

Model	CMA-GFS	CMA-MESO	ECMWF	NCEP-GFS	CMA-SH9
Evaluation score	3.30	5.50	4.61	3.55	5.41

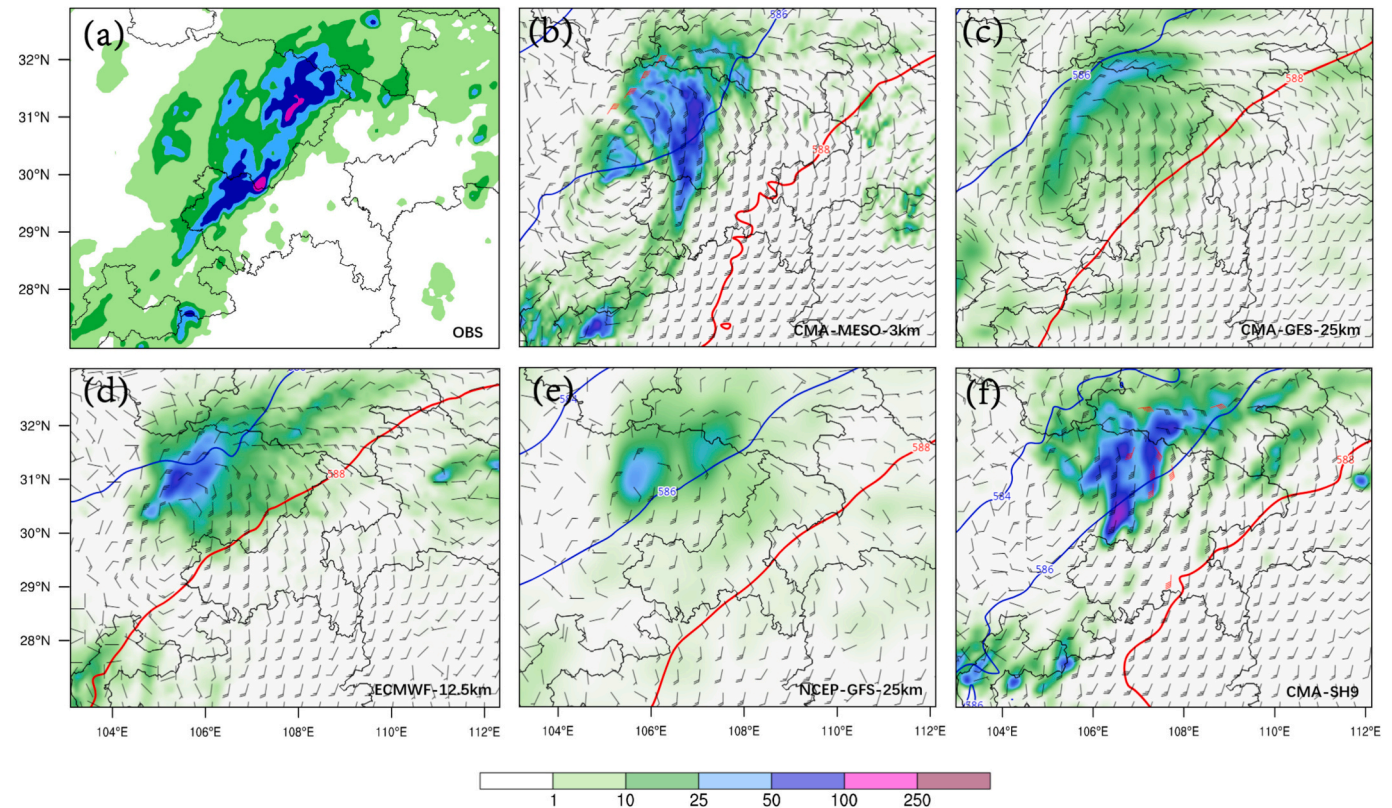


Fig. 12. Comparison of “Sep 04” heavy rainfall case, Observation and multimodal precipitation (color shadow, unit: mm), 500 hPa height field (contour, unit: gpm) and 850 hPa wind field (wind vane) forecast from 20:00 on Sep 4 to 08:00 on Sep 5, 2021: (a) Observation precipitation, (b) CMA-MESO-3 km, (c) CMA-GFS-25 km, (d) ECMWF-12.5 km, (e) NCEP-GFS-25 km combined reflectivity forecast, (f) CMA-SH9. (Note: The model initial time is all at 08:00 on Sep 4, 2021)

score averaging across all forecasters provides robust assessment metrics.

Table 4 presents the average synoptic evaluation scores from forecasters' perspectives for the 22 cases studied. The CMA-MESO model demonstrates significant reference value during the main flood season, exhibiting accurate forecasts for southwest vortex positions and convective initiation patterns. Notably, under cold air influence, its performance for heavy rainfall events (≥ 25 mm/24 h) surpasses that of global-scale models. The CMA-SH9 model maintains relatively stable performance during spring convective processes with higher uncertainty. It most accurately captures the shear line position at the convergence zone of northerly and southerly winds, and the initiation intensity of convective systems. Its southerly jet intensity is generally stronger than ECMWF but weaker than CMA-MESO, while precipitation magnitude forecasts are more moderate and better aligned with observations. The ECMWF model remains the preferred choice for forecasting light-to-moderate precipitation. However, during severe weather events, it exhibits limitations in predicting convection triggered by mesoscale weather systems, fails to provide refined forecasts, and its predictive capability for heavy rain and torrential rain events is inferior to that of mesoscale models. Additionally, the subjective evaluation reveals several characteristic issues: CMA-MESO tends to over-forecast spring heavy precipitation events and overestimate wind fields, CMA-GFS shows pronounced under-forecasting tendencies, NCEP-GFS exhibits an elevated false alarm ratio (FAR) for light-to-moderate precipitation events during the main flood season, reaches 0.4–0.5 for 10–25 mm/24 h precipitation forecasts during June–August operational periods.

4. Conclusions and discussion

This study evaluates the forecast performance of five operational models in the Eastern Tibetan Plateau region during the 2021 flood season, including both conventional objective verification metrics and subjective synoptic assessments by forecasters. The analysis is conducted seasonally, monthly, and based on dominant weather systems. Key findings include:

1. The ECMWF model demonstrates superior stability and skill in general weather conditions and light-to-moderate precipitation forecasting, however, it underperforms in heavy rainfall prediction compared to mesoscale models, particularly within the first 12 h. Mesoscale models with 3D radar data assimilation and cloud analysis systems show clear advantages in convective cloud characterization and precipitation scoring under complex orographic conditions. NCEP-GFS exhibits an elevated false alarm ratio (FAR) for light-to-moderate precipitation events during the main flood season.
2. Over the eastern Tibetan Plateau, after implementing an ECMWF-based upgrade to its land data assimilation system, CMA-MESO, compared to its global vision, achieved a considerable reduction in surface temperature RMSE. Therefore, higher-quality model initial conditions and more appropriate data assimilation methods yield more substantial improvements in forecast accuracy over complex terrain.
3. CMA-SH9 providing more information than ECMWF in terms of both convective initiation positioning and precipitation intensity prediction. While CMA-MESO exhibits good skill in capturing meso- γ scale convective features, its operational utility is limited by elevated false alarm rates, which negatively impact precipitation skill scores. Synoptic analysis of SWV cases reveals that CMA-MESO consistently overestimates wind field intensity compared to observations and other models as forecast lead time increases, with similar over-predictions in convective cloud cluster intensity and areal coverage. In contrast, CMA-SH9 shows closer alignment with observations, likely attributable to differences in model physical parameterizations and dynamic frameworks.
4. During heavy rainfall cases in the main flood season, CMA-MESO demonstrates superior performance compared to other models in forecasting convective initiation associated with cold front systems or SWVs accompanied by cold air intrusions. The model more accurately predicts both the triggering locations of convection and the corresponding radar echo patterns, showing better agreement with observations. Thus, forecasters consistently rate CMA-MESO as the most valuable model during frontal system impacts, based on Synoptic assessment.
5. For future model application and evaluation efforts, comprehensive retrospective analysis should be strengthened with timely attention to model upgrade information. While synoptic research can delve deeply into case studies, operational model verification should adopt a more holistic perspective encompassing annual performance characteristics. This approach will enable alignment with evolving climatic features of heavy rainfall and severe convection, thereby guiding the reference standards, practical applications, and improvement priorities for numerical models.

CRediT authorship contribution statement

Zhipeng Wu: Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Yan Zhang:** Writing – review & editing, Supervision. **Chunmei Hu:** Writing – review & editing, Investigation, Formal analysis. **Chengzhi Deng:** Writing – review & editing, Methodology, Investigation, Formal analysis. **Juan Luo:** Formal analysis, Data curation. **Shenggang Wu:** Formal analysis. **Yinying Zhou:** Formal analysis, Data curation. **Hong Zhang:** Writing – review & editing, Formal analysis, Data curation. **Zhiyi Liao:** Formal analysis, Data curation. **Huan Wang:** Formal analysis, Data curation. **Yue Pang:** Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.atmosres.2025.108505>.

Data availability

Data will be made available on request.

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